

Labelled finite dynamics: exact balance, projection and root adjoint

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1. Purpose and scope

This note concerns the prospective PAH-001-v2 revision 0.2.0-draft.2, not an interpretation of the immutable v1 source. The new integer counting-state and directed-label conventions were explicitly approved and frozen before the checks. The model file is [strategy/pa-hyp/PAH-001-v2-r2.json](#), with SHA-256

2e1f5f21796a224f80141572dd9dd6451dd2cbba86233998ea992aa2bff6e36a

Its source pins fix the original functional, per-root rates, mobility, external stochastic time and limit order. None is fitted or altered here. R-570 is auxiliary to T-054; the host remains T1 with its physical signature gate open.

2. Statement and setting

Fix any admissible finite anchored oriented two-cell complex and any declared finite regulator. Faces carry nonempty closed oriented boundary words, as required for the displayed boundary average J_p to be defined. Automorphisms preserve the full cell/attaching and anchor data, not just graph adjacency. Set $K \geq 2$, $M_s, M_\psi \geq 1$ integer and $0 \leq Q \leq |V|M_\psi$ integer. All real parameters have the source domain, in particular $0 < \epsilon \leq 1$, $\beta, \nu > 0$; m^2, λ_4 may be negative. The state space X consists of the distinct integer tuples

$$j_v \in \{0, \dots, M_s\}, \quad \ell_v \in \{0, \dots, M_\psi\}, \quad n_v, u_e \in \mathbb{Z}_K, \quad \sum_v \ell_v = Q.$$

Keep phase labels at $\ell_v = 0$ and aperture indices at $\epsilon = 1$. Displayed fields are

$$s_v = \epsilon + \frac{j_v(1 - \epsilon)}{M_s}, \quad \psi_v = \frac{R_{\max} \ell_v}{M_\psi} \zeta_K^{n_v}, \quad U_e = \zeta_K^{u_e}.$$

The displayed finite functional F is incorporated verbatim by the revision's parent hash and pointer. No quotient of counting states is taken.

For each primitive cell/type/sign root retain the incidence (x, r) precisely when $rx \in X$. PH and LK add the sign modulo K to n_v and u_e ; AP adds the sign to j_v without wrapping. TR along $e = (v, w)$ adds $\sigma(\delta_w - \delta_v)$ to occupations and keeps all other coordinates. Different roots remain distinct, including coincident signs at $K = 2$. Invalid moves are omitted, and identity rows are allowed.

Let $\mathcal{R}$ be the finite set of valid labelled incidences and write

$$\begin{aligned} Z &= \sum_{x \in X} e^{-\beta F(x)}, & \pi(x) &= Z^{-1} e^{-\beta F(x)}, \\ c_r(x) &= m_r(x) e^{-\beta(F(rx) - F(x))/2}, & Lf(x) &= \sum_{r: (x, r) \in \mathcal{R}} c_r(x) (f(rx) - f(x)), \\ Bf(x, r) &= \sqrt{c_r(x)/2} (f(rx) - f(x)). \end{aligned}$$

Use $D = \mathbb{C}^X = L^2(\pi)$ and the full root space $\mathbb{C}^{\mathcal{R}}$ with weight $\pi(x)$ on every directed incidence. Inner products are conjugate-linear in their first entries.

Theorem 1. *For every such finite instance, all retained roots have valid inverses, $L1 = 0$, and $\pi(x)c_r(x) = \pi(rx)c_{r^{-1}}(rx)$. The declared gauge/anchor-automorphism group average P_{cand} is self-adjoint and idempotent and satisfies $P_{\text{cand}}L = LP_{\text{cand}}$. For the actual adjoint under the displayed pairings, $B^*B = -L$ on all D . The simultaneous invariant finite subspace is preserved by L and the corresponding root lift. No cross-regulator common space is asserted.*

3. Proof: inverse maps and Gibbs balance

X is finite and nonempty. Positive s_v and nonempty finite face words make $F(x)$ finite and real, even with negative mass/quartic coefficients. Thus $0 < Z < \infty$ and every $\pi(x) > 0$. All displayed operators have full finite domains; no extension or closability inference is needed.

PH/LK inverses subtract the same residue. AP reverses its integer increment. TR subtracts its incidence vector and preserves Q ; for a loop the vector is zero. Since the original tuple lies in X , the reverse of any valid move is valid. Retained phases prevent information loss at zero occupation. Hence $\iota(x, r) = (rx, r^{-1})$ is an involutive bijection of $\mathcal{R}$.

The mobilities satisfy inverse symmetry by their exact definitions: PH uses s_v^ν ; TR/LK use $(s_v s_w)^{\nu/2}$ and leave aperture unchanged; AP uses the symmetric product of its before/after aperture values. Therefore, with $y = rx$ and $m = m_r(x) = m_{r^{-1}}(y)$,

$$\pi(x)c_r(x) = \frac{m}{Z} e^{-\beta(F(x)+F(y))/2} = \pi(y)c_{r^{-1}}(y).$$

Summing over all labels gives state-level detailed balance, without dividing by multiplicity. Also $L1 = 0$ term by term. Bare root measure $\pi(x)$ need not be preserved by ι ; only the conductance $\pi(x)c_r(x)$ is paired.

4. Proof: symmetry and projection

The gauge action sends n_v to $n_v + g_v$ and u_{vw} to $u_{vw} + g_w - g_v$. It is a bijection on tuples, not merely displayed values. The edge term transforms by

$$\psi'_w - U'_{vw}\psi'_v = \zeta_K^{g_w}(\psi_w - U_{vw}\psi_v).$$

Its absolute square is unchanged; onsite and aperture terms are invariant. Gauge factors telescope on every closed face word, proving exact F and π invariance. All four partial moves commute with this action; their validity and mobility are unchanged. Thus their rates are equivariant too.

An anchor-preserving cell automorphism permutes vertex coordinates. If it sends stored edge e to e' with orientation sign τ_e , it sends u_e to $\tau_e u_e$ and $\text{TR/LK}(e, \sigma)$ to $\text{TR/LK}(e', \tau_e \sigma)$. PH/AP signs do not change. This exactly intertwines the partial moves and preserves their mobilities. Reversed edge energy is unchanged because

$$|\psi_v - U_e^{-1}\psi_w|^2 = | -U_e^{-1}(\psi_w - U_e\psi_v)|^2 = |\psi_w - U_e\psi_v|^2.$$

Reversal conjugates a unit face holonomy, preserving its real part; boundary averages and multiplicities are permuted. Hence F, π and rates are invariant.

For either group, the pullbacks $U_h f(x) = f(h^{-1}x)$ are unitary. Root reindexing gives $U_h L = L U_h$. The finite average P_H is self-adjoint by inversion and idempotent because each group product occurs $|H|$ times. Automorphisms normalize the gauge group by permuting its vertex coordinates; therefore $P_{\text{Aut}} P_G = P_G P_{\text{Aut}}$. Individual gauge and automorphism actions need not commute. The product P_{cand} is consequently an orthogonal projection commuting with L , whose range is invariant.

5. Proof: the weighted adjoint

Expand $\langle Bf, h \rangle_{\mathcal{R}}$ and collect each $\overline{f(x)}$. Because all sums are finite and $\pi(x) > 0$, the adjoint is

$$\begin{aligned} B^*h(x) &= \frac{1}{\pi(x)} \sum_{(y,r): ry=x} \pi(y) \sqrt{c_r(y)/2} h(y, r) \\ &\quad - \sum_{r:(x,r) \in \mathcal{R}} \sqrt{c_r(x)/2} h(x, r). \end{aligned}$$

Substitute $h = Bf$. The incoming part equals $\frac{1}{2} \sum_r c_r(x)(f(x) - f(rx))$ by the incidence bijection and detailed balance. The outgoing part is the same expression, giving $B^*B = -L$. Identity rows contribute zero. No extra rate or half weight is put in the root measure. Equivalently, pairing directed incidences proves

$$\frac{1}{2} \sum_{(x,r)} \pi(x) c_r(x) \overline{f(rx)} - \overline{f(x)} (g(rx) - g(x)) = -\langle f, Lg \rangle_{\pi}.$$

This is the full sesquilinear identity, not a real-observable sample test. The root symmetry pullback is unitary and intertwines B with U_h . Thus the identity also restricts to the candidate invariant spaces.

6. Numerical verification and kernel scope

The preregistration and detailed certificate are [strategy/pa-hyp/PAH-v2-finite-audit-v1.json](#) and [strategy/pa-hyp/PAH-v2-finite-proof-v1.md](#). The fixture is an anchored triangle with $K = 2$, $M_s = M_\psi = 1$, $Q = 1$, $\epsilon = 1/2$, $\nu = 2$; all remaining exact inputs are in the preregistration. It is an instance of the unchanged model, not the universal proof.

Primary verification uses rational coefficients of formal exponential monomials and independently assembles the weighted generator and root Gram matrices. It checks 1536 states and 26112 valid incidences, with exact coefficient equality. A non-importing flattened-state implementation forms the full finite matrices and orbit projection independently; the maximum reported absolute residual is below $5.0 \cdot 10^{-15}$, with tooling tolerances `rtol=1e-11`, `atol=1e-14`. This floating comparison is corroboration, not an exact proof or a uniform error estimate.

Twelve hostile controls detect coefficient, root-measure, label, endpoint, gauge-sign, orientation and projection shortcuts. Lean 4.32.1 compiles nine parameterized declarations for inverse arithmetic, conservation, Gibbs flux, the square-root half factor, complex pair algebra and commuting idempotents. The dependency revisions are checked. Lean does not encode the complete cell-complex model or the full symmetry/group-average instantiation. All scripts were authored in this task; implementation independence is not an external-person mathematical audit. External review remains invited.

Run each `pah_v2_finite_primary.py`, `pah_v2_finite_independent.py`, `pah_v2_finite_hostile.py`, and `pah_v2_finite_lean.py` under [verification/scripts/](#), with `python -X utf8` and `-check`. The Lean runner additionally accepts `-lean-cache` pointing to the pinned cached packages. Runs are under [claims/C6-SPACETIME-SIGNATURE/runs/2026-09-11-pah-v2-finite/](#).

7. Devil's-advocate and literature

The finite balance/Dirichlet formalism is standard background: Aldous–Fill, *Reversible Markov Chains and Random Walks on Graphs*, section 3.1, equation (3.4), and section 3.6.1, equations (3.74)–(3.76), author-hosted at <https://www.stat.berkeley.edu/~aldous/RWG/book.html>. The identities required here are rederived above; irreducibility-dependent uniqueness/mixing statements are not imported. R-479's separate successor PASS is not reused as proof of this revision.

1. Factor/measure objection: two directions might double the form. DISMISSED by the explicit two halves in the actual adjoint calculation. Bare-root reversal invariance is UPHELD as a false shortcut and not used.
2. Domain objection: zero-radius or unit-floor values might identify states and erase inverses. DISMISSED by retained integer labels. Empty face words are a VALID domain warning: an undefined J_p is not certified.
3. Symmetry objection: static invariance need not imply dynamic symmetry. UPHELD in general; exact root/mobility equivariance is separately established. Elementwise Aut-gauge commutation is not assumed.
4. Limit objection: finite identities might imply refinement or a limiting semigroup. UPHELD as overclaim. No comparison maps, uniform estimates or limits are supplied here.
5. Independence objection: computer PASS might replace a general proof or external review. UPHELD as a scope warning. Universal quantifiers are carried by the proof, and same-author executable independence is explicitly limited.

8. Result footer

Result ID:	R-570 / C6-SPACETIME-SIGNATURE
Precise statement:	Five finite operator identities for pinned v2.
Scope:	Every defined admissible finite v2 instance.
Dependencies:	Frozen v2, immutable parent formulas, approvals.
Evidence grade:	ANALYTIC, EXACT, EXECUTED; partial Lean.
Reproduction command:	See section 6; four standalone --check commands.
Expected output:	PASS at each explicitly declared check scope.
Falsification gate:	An exact counterexample to one finite identity.
Tier before / after:	Host C6 T1 / T1; no physical promotion.
No-overclaim statement:	No refinement, Pre-A, QFT, gravity or limit.
Next required action:	Separate fixed comparison-map admission.